Aurora Solar Microgrid Pilot

Field Performance Report, Version 2.0

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Executive Summary

The Aurora pilot installed solar and battery microgrids at 4 remote sites. The pilot reduced diesel consumption by 46% compared with the baseline year. Average system uptime reached 98.6%, and total generation was 468,000 kWh over the reporting period.

The estimated payback period is 5.1 years. Total capital expenditure was USD 2,320 thousand.

Key Findings

- Diesel consumption fell by 46%.
- System uptime: 98.6%.
- Payback period: 5.1 years.
- Zephyr-7 inverters performed within specification at all sites.

1. Background and Methodology

Remote communities in the region have historically relied on diesel generators, which are costly to fuel and maintain. The programme tested hybrid solar-battery microgrids to lower fuel use and improve reliability. Each site was instrumented with smart meters that logged generation, load and fuel use every fifteen minutes.

Selection Criteria

- Distance from the main grid greater than 40 km.
- Documented annual diesel use above 20 kilolitres.
- Community agreement and available land for the array.

Table 1: Pilot sites and installed capacity

Site	Terrain	Solar	Battery
Harbor Point	Coastal	140 kW	120 kWh
Ridge Station	Mountain	85 kW	64 kWh
Delta Camp	River delta	75 kW	72 kWh
Cape Outpost	Island	50 kW	40 kWh

Data validation followed the programme's two-step review: automated range checks, then manual review by an independent engineer.

1.1 Scope Expansion (new in version 2.0)

Cape Outpost, an island site, was added in the second phase to test performance in high-salinity conditions.

2. Results

Monthly diesel use dropped sharply once the arrays were commissioned. The chart below compares kilolitres of diesel consumed per month before and after installation.

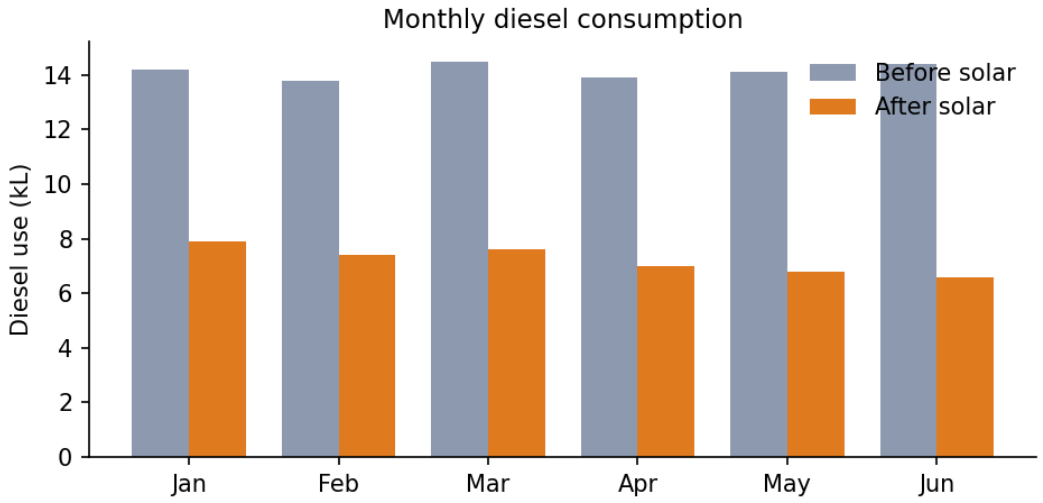

Figure 1: Monthly diesel consumption before and after solar installation, in kilolitres.

Table 2: Energy supplied by source

Month	Total load (kWh)	Solar supplied (kWh)	Solar share
January	58,400	36,700	63%
February	54,900	34,600	63%
March	61,300	39,900	65%

Solar share rose each month as operators tuned battery dispatch schedules.

3. Costs and Risks

Capital expenditure totalled USD 2,320 thousand. The storage units selected were model BATT-4410.

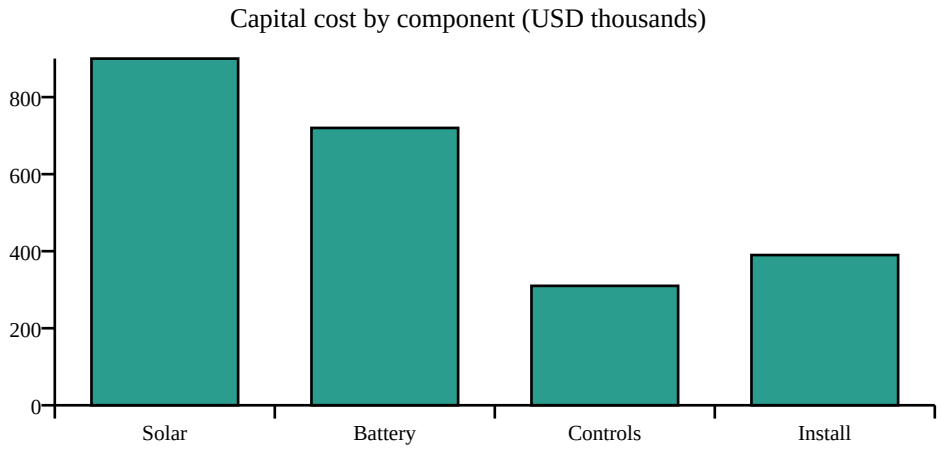

Figure 2: Capital cost breakdown by component, in USD thousands.

Table 3: Capital cost breakdown

Component	Cost (USD k)	Share
Solar array	900	39%
Battery storage	720	31%
Controls and grid	310	13%
Installation	390	17%

Risk Register

- Battery degradation above 3% per year could extend payback.
- Typhoon season may interrupt maintenance access.
- Inverter supply now dual-sourced; vendor dependency risk resolved.

4. Conclusion

Version 2.0 recommends a full rollout to six sites in 2026 and approves procurement of the BATT-4410 storage units.

Contact: energy-pilot@example.org. This report contains no information about wind generation or hydrogen storage.